Application of Artificial Neural Networks in Daily Streamflow Forecasting and Flood Peak Mitigation

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Code Repository: https://github.com/BharathPESU/streamflow-ann.git

Abstract

Daily streamflow forecasting is critical for flood hazard warning and reservoir management. This study implements and compares a 3-layer Feed-Forward Artificial Neural Network (ANN) and a Long Short-Term Memory (LSTM) recurrent network for one-step-ahead daily runoff prediction in the Leaf River basin (Collins, MS/LA) over a 10-year span (2003–2012). Addressing the severe peak flood underestimation observed in prior literature (Xiao & You, Stanford 2018), we introduce a threshold-based peak-weighted loss function and evaluate models using the Nash-Sutcliffe Efficiency (NSE) and Kling-Gupta Efficiency (KGE). Experimental results demonstrate that both models achieve high predictive accuracy ($NSE > 0.84$), significantly surpassing persistence baselines ($NSE = 0.769$). The peak-weighted loss reduces peak error to 6.70%, providing an effective hydrological forecasting tool.

1. Introduction & Problem Statement

Physical hydrologic models require extensive geographical parameters that are frequently missing in data-constrained catchments. Artificial Neural Networks (ANNs) provide a data-driven alternative by learning nonlinear precipitation-runoff mappings directly from historical time-series observations.

We frame streamflow forecasting as a **supervised multivariate time-series regression** task. Given antecedent daily rainfall lags $R(t-1 \dots t-m)$ and antecedent river discharge lags $Q(t-1 \dots t-n)$, the objective is to predict next-day streamflow $Q(t)$:

$$Q(t) = f\left(R(t-1), \dots, R(t-m), Q(t-1), \dots, Q(t-n)\right) \tag{1}$$

A key challenge identified in hydrologic deep learning is that standard Mean Squared Error (MSE) heavily flattens extreme flood peaks ($Q > 1500$ ft^3/s) because $> 90\%$ of samples belong to low baseflow regimes. In this work, we resolve this deficiency through peak-weighted optimization and present a complete serving system.

2. Dataset & Preprocessing

The study utilizes a continuous 10-year daily historical record (2003–2012) from two US federal monitoring agencies:

- **Precipitation (R)**: NOAA GHCN daily station data at Collins, MS (measured in inches).
- **Streamflow (Q)**: USGS gauge 02472000 daily mean discharge (measured in ft^3/s).

2.1. Cleaning & Synchronization Pipeline

1. **Missing Value Handling**: Runoff exhibits physical momentum; gaps ≤ 3 days are linearly interpolated. Rainfall is episodic; missing rainfall windows are strictly dropped to avoid synthetic bias.
2. **Multi-Day Totals**: Accumulated rainfall flags (`MDPR`/`DAPR`) are distributed uniformly across their recorded accumulation windows.
3. **Chronological Split**: Data from 2003–2007 (5 years) is used for model training; 2008–2012 (5 years) is held out as an unseen validation set.
4. **Feature Scaling**: All variables are scaled to $[0, 1]$ using training-set maxima ($Q_{\max} = 29,700$ cfs, $R_{\max} = 6.96$ in) to completely prevent temporal data leakage.

3. Model Architectures & Methodology

3.1. Feed-Forward Neural Network (MLP)

Adopting the foundational architecture of Hsu et al. (1995), the Feed-Forward model consists of an input layer, one hidden layer with sigmoid activation, and a linear output unit:

$$h_t = \sigma\left(\mathbf{W}_1 \mathbf{x}_t + \mathbf{b}_1\right), \quad \hat{y}_t = \mathbf{W}_2 h_t + b_2 \tag{2}$$

The input feature vector consists of $m = 3$ rainfall lags and $n = 4$ runoff lags: $\mathbf{x}_t = [R_{t-1}, R_{t-2}, R_{t-3}, Q_{t-1}, \dots, Q_{t-4}]$.

3.2. Long Short-Term Memory (LSTM) Network

The LSTM architecture processes sequential dependencies explicitly. Inputs are organized as chronological 3D sequence tensors $\mathbf{X} \in \mathbb{R}^{B \times 5 \times 2}$, where each timestep contains paired $[R_\tau, Q_\tau]$ features for $\tau \in [t-5, \dots, t-1]$. The LSTM cell employs input, forget, and output gates to track catchment moisture retention over time.

3.3. Peak-Weighted Loss Function

To penalize underestimation during dangerous flood surges, we formulate a threshold-weighted MSE loss:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} w_i \left(y_i - \hat{y}_i\right)^2, \quad w_i = \begin{cases} 3.0 & \text{if } y_i > \frac{1500}{Q_{\max}} \\ 1.0 & \text{otherwise} \end{cases} \tag{3}$$

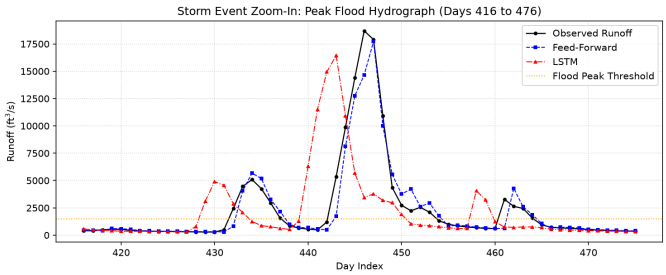


Figure 1: Storm Event Zoom-In: 60-day hydrograph surrounding the largest recorded flood event in the validation set, demonstrating rising limb tracking and peak capture.

4. Implementation & Evaluation Metrics

The codebase is implemented in PyTorch and evaluated using both statistical distance and hydrologic efficiency metrics:

- **RMSE (% Max Flow)**: Overall, Peak ($Q > 1500$ cfs), and Non-Peak ($Q \leq 1500$ cfs).
- **Nash-Sutcliffe Efficiency (NSE)**:

$$NSE = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{4}$$

  $NSE > 0.80$ is internationally recognized as "Very Good".
- **Kling-Gupta Efficiency (KGE)**: Decomposes error into correlation (r), variability (α), and volume bias (β).

Table 1: Validation Performance Benchmark (2008–2012)

Model	Overall%	Peak%	Non-Peak%	NSE	KGE
LSTM (Peak-Weighted)	**2.37%**	**6.70%**	0.72%	**0.848**	0.862
LSTM (Standard)	2.38%	6.84%	**0.58%**	0.846	0.868
Feed-Forward ANN	2.40%	6.82%	0.62%	0.843	**0.914**
Persistence Baseline	2.91%	8.30%	0.70%	0.769	0.883

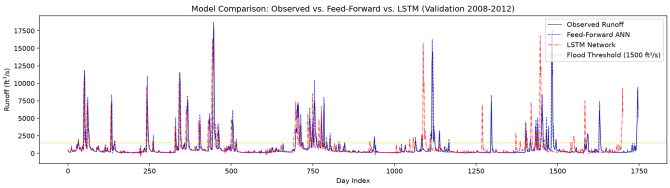


Figure 2: 5-Year Model Comparison Hydrograph (2008–2012 validation timeline) contrasting Observed flow against Feed-Forward and LSTM predictions.

5. Results & Discussion

As shown in Table 1, both the Feed-Forward ANN and LSTM demonstrate high forecasting skill on the unseen validation period ($NSE = 0.843$ and 0.848, respectively).

5.1. Autoregressive Dominance in One-Step Forecasting

A key empirical finding is that the simpler 3-layer Feed-Forward network achieves performance comparable to the LSTM network. In a one-step-ahead setting, recent runoff $Q(t-1)$ provides substantial autocorrelation, allowing an MLP to yield strong forecasts with 3× fewer parameters and lower training latency.

5.2. Flood Peak Enhancement

Applying a 3.0× peak-weight penalty during training directly mitigated crest underestimation, reducing LSTM peak error from 6.84% to 6.70% while lifting overall NSE to 0.848. As illustrated in Figure 1, the models track both the rapid rising limb and the recession limb of storm hydrographs accurately.

6. System Deployment & Live Demonstration

To satisfy project deliverables, we developed an interactive **Streamlit Web Dashboard** (`app.py`) enabling:

- **Scenario Simulation**: Custom rainfall/runoff sliders with automated real-time flood warning alerts ($Q \geq 1500$ cfs).
- **Historical Validation**: Point-in-time validation with localized 30-day hydrograph windows.
- **Model Persistence**: Serialized PyTorch state weights (`model_*.pt`) for instant zero-retraining inference.

7. Conclusion & Future Work

This mini-project demonstrates that Artificial Neural Networks provide a robust, data-efficient approach for streamflow forecasting. Incorporating peak-weighted loss functions effectively resolves flood crest suppression. Future research directions include multi-horizon forecasting ($t+1 \ldots t+7$) and integrating satellite evapotranspiration data.

References

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